

Introduction to the World of AI and Data Science

Agenda

- Decision-Making
 - Quantitative Vs Qualitative
 - Learning from Data
- Artificial Intelligence
 - What it is?
 - A brief history

Intelligent machines!



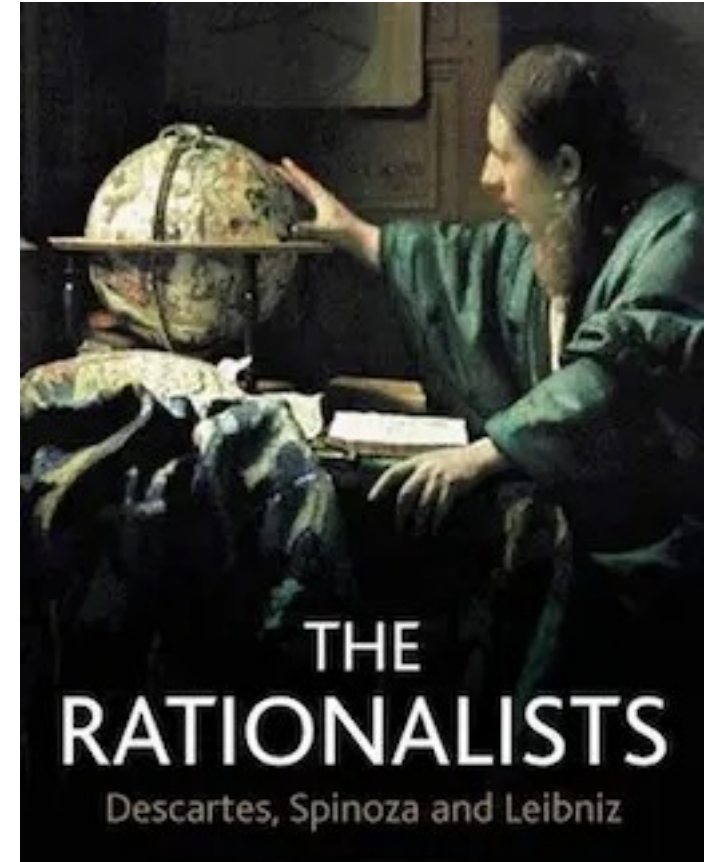
Greek Mythology

Talos - bronze automaton
(3rd BCE)



Mechanical Devices

Al-Jazari's mechanical
humanoid (1200)

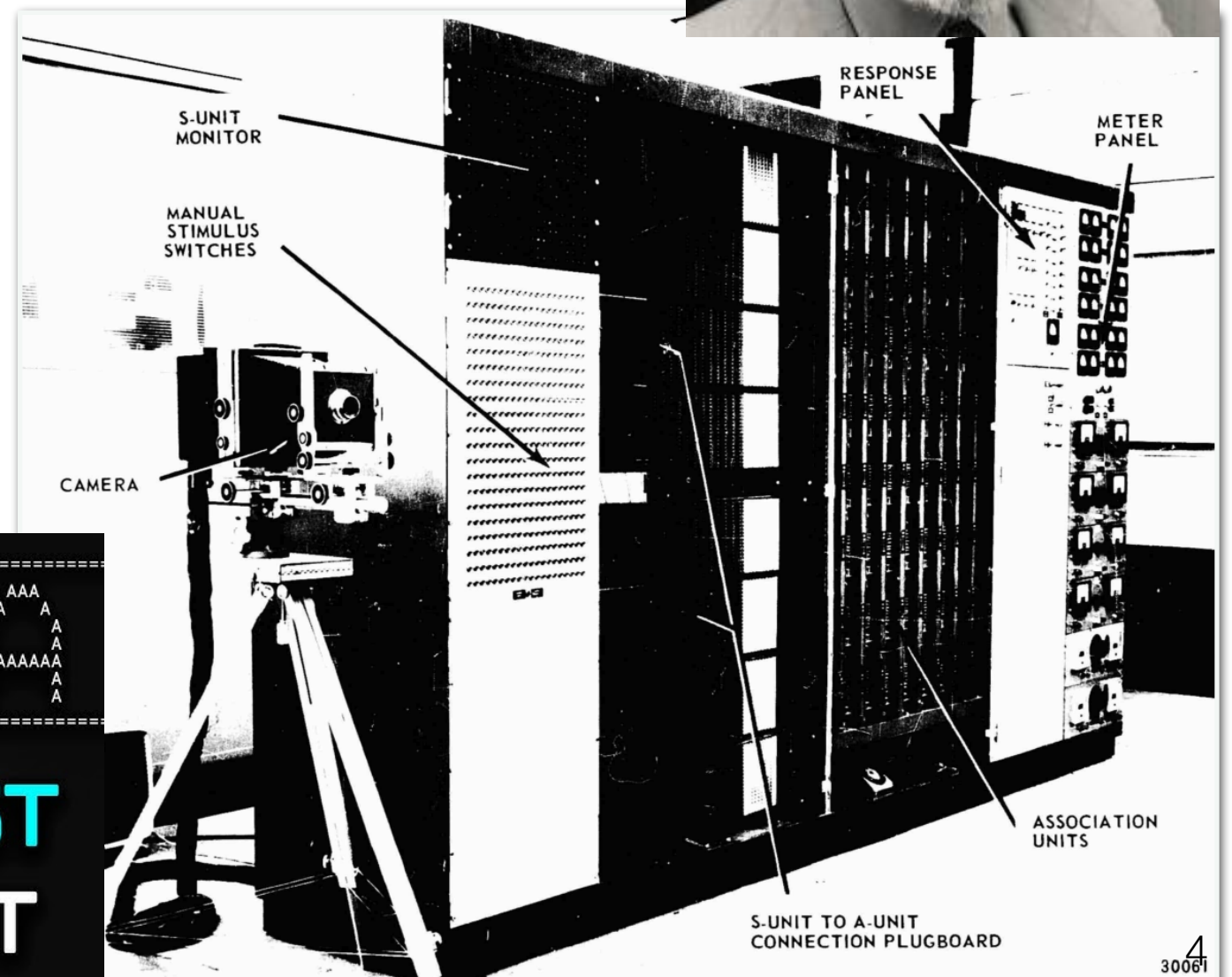


Philosophy

Brain, Intelligence and Math
(1600)

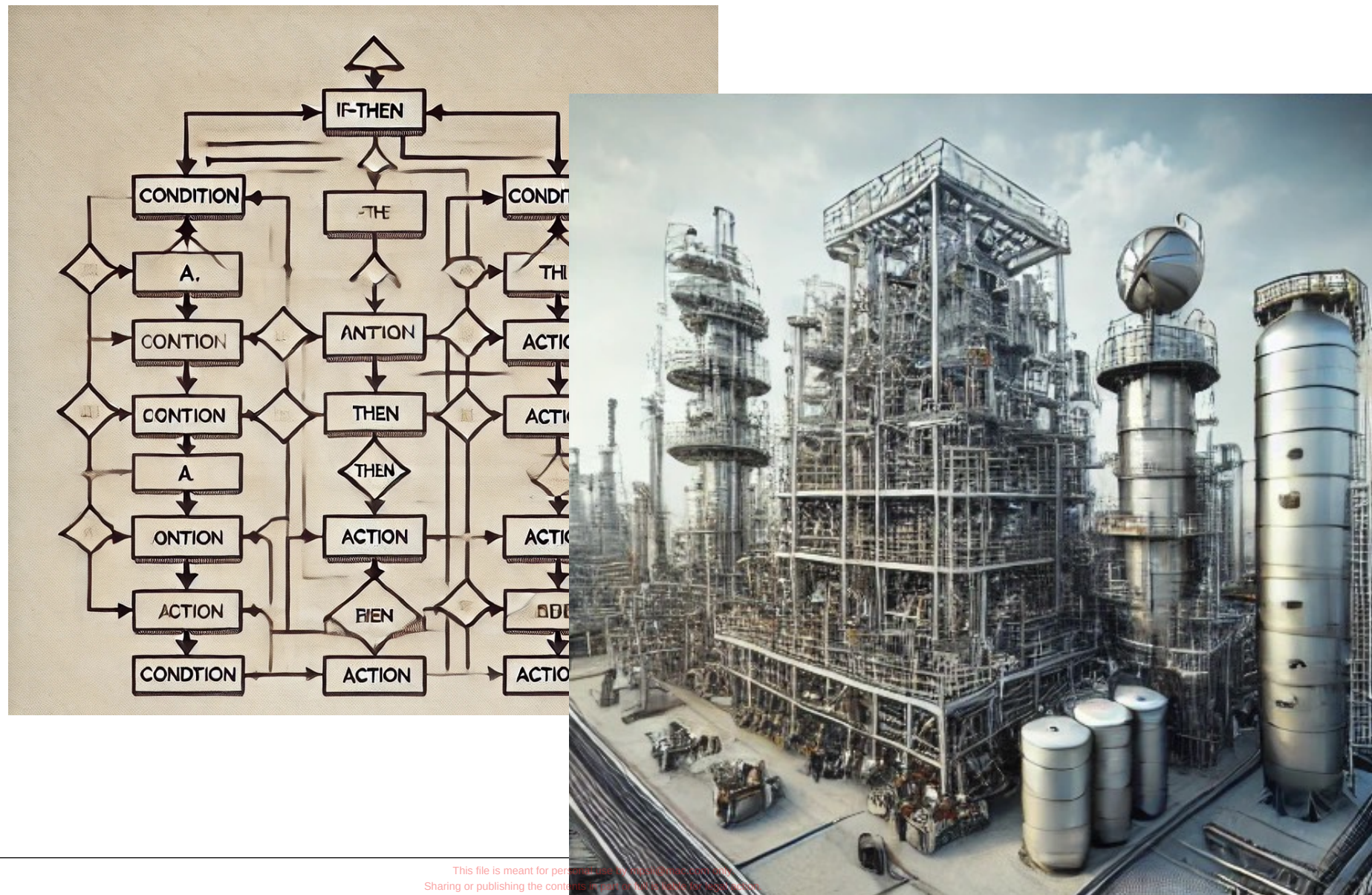
Early days

- McCulloch-Pitts Neuron, 1940
 - Alan Turing (1950): The Turing test
 - John McCarthy (1956): Artificial Intelligence
 - “the embryo of an electronic computer that [the Navy] expects will be able to walk, talk, see, write, reproduce itself and be conscious of its existence.” - Rosenblatt, (1958)
- 



before LLMs!

- Symbolic AI (1950s - 1980s), Expert Systems (1970s - 1990s)
- Methods: Rule-based systems, knowledge representation.
- Applications: Medical diagnosis systems, financial advisory tools. Expert systems like MYCIN (diagnostic system), Dendral (chemical analysis), and SHRDLU (natural language processing).



before LLMs!

- Dynamic programming and Reinforcement Learning (1990s - abt 2020!)
- Methods: Dynamic Programming, Stochastic Control, Markov decision processes, Q-learning, deep reinforcement learning.
- Applications: Game playing (e.g., Deepblue, AlphaGo), recommendation systems, Autonomous robotics.



before LLMs!

- Neural Networks (1940s - 1960s, resurgence in the 2010s)
 - Methods: Perceptrons, multi-layer perceptrons, deep learning.
 - Applications: Image and speech recognition, Generative AI
 - Computer Vision and Robotics: Autonomous vehicles, industrial automation, medical image analysis.
 - Natural Language Processing (NLP): Chatbots, language translation (e.g., Google Translate), sentiment analysis.
 - Generative AI : GPTs, Deepfakes, Music etc.

